TEAM EAGLE SHIELD

Team 25

University of Prishtina

INTRUSION REPORT

Iowa Cyber Defense Competition (ICDC) 2026

Spring 2026

Table of Contents

1. Executive Summary

2. Server-by-Server Analysis

1. 2.1 Active Directory
2. 2.2 Ticketing Machine
3. 2.3 Management Backend
4. 2.4 Admin UI
5. 2.5 Public Website

3. Summary

1. Executive Summary

Throughout the ICDC 2026 competition, Team Eagle Shield actively monitored all servers for signs of intrusion using Windows Event Viewer, system logs (journalctl), firewall logs, and network monitoring tools. This report documents the security incidents detected during the attack phase.

Key Findings

Total Flags Lost	**6**
Servers Secured	2 (Ticketing, Admin UI)
Servers Compromised	2 (Management Backend, Public Website)
Servers Partially Compromised	1 (Active Directory)

2. Server-by-Server Analysis

2.1 Active Directory Server

IP Address	108.64.203.10
Status	PARTIALLY COMPROMISED
Flags Captured by Red Team	1
Monitoring Tool	Windows Event Viewer

Findings: The Active Directory server was partially compromised. Red Team successfully gained Administrator-level access to the C:\Users\Administrator directory (red flag captured). However, they were unable to gain SYSTEM-level privileges as the blue flag in C:\Windows\System32 remains secure. We detected multiple RDP brute force attempts (Event ID 4625) throughout the attack phase. Analysis of Event Viewer logs shows suspicious successful logon events (Event ID 4624) from external IP addresses, indicating Red Team gained admin access, likely through credential theft, privilege escalation, or exploitation of an Active Directory vulnerability.

Evidence: Windows Security Event Log showing Event ID 4624 (Successful Logon) from Red Team IP. Red flag present in C:\Users\Administrator directory. Blue flag in C:\Windows\System32 still secure. Event ID 4625 entries showing earlier failed attempts before successful breach.

Impact: MEDIUM - Red Team gained administrative access but not SYSTEM privileges. Domain functionality continued. All machines remained joined to the domain. LDAP/LDAPS services operational.

2.2 Ticketing Machine

IP Address	108.64.203.20
Status	SECURE
Flags Captured by Red Team	0
Monitoring Tool	Windows Event Viewer, Application Logs

Findings: The Ticketing Machine server remained completely secure throughout the competition. All flags (Blue: C:\Windows\System32, Red: C:\Users\Administrator, and Forged Ticket) remain unclaimed. We detected reconnaissance activity and connection

attempts to port 5000 (HTTP service), but no successful compromise was achieved. Strong passwords and proper security configuration prevented Red Team access. The ticketing application continued to function properly, generating and validating tickets without interruption.

Evidence: Dashboard showing all Ticketing flags as "Unclaimed". Windows Event Viewer showing only Event ID 4625 (Failed Logons), no successful unauthorized access. Application logs showing normal ticket operations.

Impact: NONE - Server maintained complete security. No compromise detected.

2.3 Management Backend

IP Address	108.64.203.30
Status	COMPROMISED
Flags Captured by Red Team	3
Monitoring Tool	journalctl, /var/log/auth.log

Findings: The Management Backend server was fully compromised. Red Team successfully gained root access (red flag in /root/ captured) and performed multiple malicious actions: (1) Database Access - captured "MGMT DB Read" flag by extracting sensitive payment data from the database, (2) Service Disruption - disabled all buses by capturing "MGMT Bus service" flag, (3) Service Disruption - disabled all trains by capturing "MGMT Train service" flag. However, the blue flag in /etc/ remains secure, indicating Red Team did not modify system configuration files. Analysis of auth.log and journalctl revealed successful SSH access from Red Team IPs after initial failed attempts, suggesting eventual password compromise or exploitation.

Evidence: /var/log/auth.log showing "Accepted password for root" from Red Team IP. journalctl showing database queries accessing payment records. Application logs showing API calls to /vehicles/<bus>/flag and /vehicles/<train>/flag endpoints. Red flag confirmed in /root/ directory. Blue flag in /etc/ still secure.

Impact: **CRITICAL** - Complete system compromise. Root access achieved. Database breached with potential exposure of payment information and user data. Public

transportation service disrupted with all buses and trains disabled. Financial and operational impact severe.

2.4 Admin UI

IP Address	108.64.203.40
Status	SECURE
Flags Captured by Red Team	0
Monitoring Tool	Windows Event Viewer, Firewall Logs

Findings: The Admin UI server remained completely secure despite experiencing a temporary firewall misconfiguration. All flags (Blue: C:\Windows\System32, Red: C:\Users\Administrator) remain unclaimed. The firewall issue (OFF state with BlockInbound policy) was quickly identified and resolved by enabling proper firewall rules. Multiple RDP brute force attempts were detected and blocked. Our proactive security measures including strong passwords, proper firewall configuration, and rapid incident response prevented any successful compromise. The Admin UI maintained functionality after the firewall fix, allowing administrators to monitor the transportation fleet.

Evidence: Dashboard showing all Admin UI flags as "Unclaimed". Windows Event Viewer showing only Event ID 4625 (Failed Logons). netstat confirming application listening on 0.0.0.0:80. Test-NetConnection results showing TcpTestSucceeded : True after firewall remediation. No unauthorized Event ID 4624 entries detected.

Impact: NONE - Server maintained complete security. Brief service unavailability due to firewall misconfiguration but no compromise occurred.

2.5 Public Website

IP Address	108.64.203.50
Status	COMPROMISED
Flags Captured by Red Team	1
Monitoring Tool	journalctl, HTTP logs, SSH logs

Findings: The Public Website server was compromised. Red Team successfully gained root access (red flag in /root/ captured). However, the blue flag in /etc/ remains secure, indicating system configuration files were not accessed or modified. HTTP access logs show extensive web application probing including SQL injection attempts, XSS probes, and directory traversal attacks. SSH authentication logs show numerous failed login attempts followed by eventual successful access, indicating password compromise or SSH vulnerability exploitation. Despite root compromise, the website continued to function, suggesting Red Team focused on flag capture rather than service disruption.

Evidence: /var/log/auth.log showing "Accepted password for root" or "Accepted publickey for root" from Red Team IP. journalctl confirming privilege escalation to root. Red flag present in /root/ directory. Blue flag in /etc/ still secure. HTTP access logs showing SQL injection patterns (OR 1=1, UNION SELECT) and suspicious requests.

Impact: **HIGH** - Root access compromised on public-facing server. Potential exposure of citizen data (accounts, ticket purchases, personal information). However, service continued operating, minimizing impact to users. Database integrity unknown pending further analysis.

3. Summary

The following table provides a complete summary of all servers and their security status during the ICDC 2026 competition.

Server	Flags Lost	Status
Active Directory	1	PARTIALLY COMPROMISED
Ticketing Machine	0	**SECURE**
Management Backend	3	COMPROMISED
Admin UI	0	**SECURE**
Public Website	1	COMPROMISED
TOTAL	6	---

Conclusion: Team Eagle Shield defended 2 out of 5 servers (Ticketing Machine and Admin UI) completely during the ICDC 2026 competition. The Management Backend and Public Website servers were fully compromised with root access gained and a total of 4 flags captured from these systems. Active Directory was partially compromised with administrative but not SYSTEM privileges obtained. Total flags lost: 6 out of 14 available flags. Despite the compromises, critical services remained operational. The ticketing system continued to function, allowing citizens to purchase and use tickets. Our rapid response to the Admin UI firewall issue prevented that server from being compromised. Strong security measures on the Ticketing Machine proved effective against all Red Team attacks.

What Worked Well:

- • Ticketing Machine remained completely secure - strong password policies effective
- • Admin UI protected through rapid firewall remediation
- • Active Directory maintained domain functionality despite partial compromise
- • Blue flags in /etc/ and System32 directories remained secure on all systems
- • Continuous monitoring enabled detection of intrusions

Areas for Improvement:

- • Management Backend was completely compromised - needs stronger access controls
- • Public Website vulnerable to attack - web application security insufficient

- • SSH/RDP password policies need strengthening across all systems
- • Need intrusion prevention system (IPS) not just detection (IDS)
- • Database access controls require hardening
- • Should implement IP whitelisting for administrative access